

Planetary Data System archive

Cassini-VIMS global hyperspectral mosaics and phase functions for Saturn's satellites Dione and Rhea

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Document review

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Data Selection

We describe here the steps performed to produce the high-quality hyperspectral global mosaics of Cassini's Visual and Infrared Mapping Spectrometer (VIMS) spectral data for Saturn's icy satellites Dione and Rhea.

VIMS was made up of two slit spectrometers, or channels, covering the visual (VIMS-V) and infrared (VIMS-IR) spectral ranges at 0.35-1.05 μm and 0.88-5.12 μm and an average spectral sampling of 7.3 and 16.6 nm, respectively (*Brown et al., 2004; McCord et al., 2004*). Each VIMS data file is three-dimensional, with two spatial dimensions ("samples", along the horizontal direction; "lines", along the vertical direction) and one spectral dimension, and it is commonly referred to as a hyperspectral "cube".

The mosaics were created by using the data from VIMS-IR only.

The first step to produce mosaics of VIMS-IR cubes was the selection of the best and most complete database for each satellite.

The data selected have a phase angle range of 10°-50°, and they included all cubes available to be downloaded from the NASA PDS Image Atlas from 2004 to 2015.

Data calibration

The data downloaded from the NASA PDS Image Atlas were calibrated by using the following USGS's Integrated Software for Imagers and Spectrometers (ISIS) routines:

1. *VIMS2ISIS*: This ISIS routine translates raw VIMS cubes into ISIS4 standard format and separates each cube into two cubes, one containing the data from the infrared channel, and one

containing the data from the visual channel.

2. *SPICEINIT*: This ISIS routine creates a camera model for each cube by searching for the appropriate information about spacecraft position, pointing, body shape and orientation, sun position, and time which are required for ground positions (latitude/longitude) and photometric viewing angles computation. This data is known as the SPICE kernels, from the NASA's Navigation and Ancillary Information Facility (NAIF), which is also publicly available.

3. *VIMSCAL*: This ISIS routine applies the most appropriate radiometric correction to the VIMS cubes. First, it applies a dark current and a flat field correction to images, then converts the output units to I/F. The to-date version of *VIMSCAL* uses the radiometric correction *RC19*, which corrects the wavelengths time-shift affecting VIMS cubes (*Clark et al. 2018*).

Saturation removal and noise attenuation

The latest version of *VIMSCAL* automatically removes the saturation effect (which involves the portion of the spectrum 0.9-2.9 μm) by deleting the spectral channels affected by this issue. ISIS, though, doesn't currently provide any program or pipeline dedicated to noise correction. To create high-quality mosaics of VIMS cubes, the spectral channels showing very low signal strength must be eliminated by manually selecting only the portion of the spectrum unaffected by noise. This task was performed by using the ENVI (Environment for Image Visualization by L3Harris Geospatial) program to open each cube and evaluate for each pixel the portions of the spectrum to retain and to discard.

Photometric correction

Combining spectra from different VIMS-IR observations photometric effects due to the variability of illumination and geometry conditions. Therefore, a photometric correction was applied to the VIMS-IR cubes of each satellite by using a code written in IDL (Interactive Data Language by L3Harris Geospatial) language.

The photometric correction first accounts for the effects of the large-scale geometry of illumination and observation due to the curvature of the satellite. A simple correction of the incidence angle such as the division by the cosine of the solar incidence angle (Lambertian model) is not adequate for the satellites of Saturn because icy planetary surfaces do not usually scatter light isotopically. The disk-function developed by *Akimov (1975, 1980, 1988)* as shown in *Shkuratov et al. (1999; equation 29)* was applied. The equation has no free term parameter, but it only depends on the incidence, emergence and phase angles.

For the saturnian satellites, the most accurate shape models that are available in the kernels are ellipsoids. Photometric calculations based on ellipsoids still enable accurate phase function corrections, because the phase does not depend on the topography.

After the disk function correction, the dispersion is reduced, and most of the remaining effects are due to the physical structure of the surface materials at scales smaller than a pixel, which change the apparent brightness as function of the phase angle.

To further correct the effects induced by photometry, a phase function was applied. For each single VIMS-IR wavelength, the I/F value of each pixel of each cube was plotted against the corresponding phase angle value. The data were fitted with a second-degree polynomial, and the phase effect was corrected by dividing the data by the best-fit curve. The polynomial parameters calculated for each wavelength are listed in the *Satellite_name>_phase_function_parameters.txt*

file. The entire photometric correction was finally setup to normalize the photometry at a geometry equivalent to 30° incidence and 0° emergence.

Geometry calculation

The geometric information necessary to project each VIMS-IR hyperspectral cube were calculated by applying a set of routines, written in IDL language, to navigate within each VIMS cube based on the use of publicly available **Navigation and Ancillary Information Facility (NAIF)’s SPICE libraries and kernels**. The routines calculate all the geometric information useful for cube projection (planetocentric and planetographic longitude and latitude, and illumination angles of the pixel’s center; planetocentric and planetographic longitude and latitude, and illumination angles of sixteen points of the pixel’s perimeter) independently for every VIMS pixel.

Creation of the mosaics

VIMS-IR hyperspectral cubes were combined together to create two hyperspectral mosaics for each satellite.

- **Optimized signal-to-noise ratio:** the cubes of each dataset were averaged together where they spatially overlap. The advantage of having this kind of mosaic is to further minimize the noise affecting the original spectra.
- **Optimized spatial resolution:** the cubes of each dataset were weighted averaged together where they spatially overlap, by using the resolution value before re-sampling for the projection as weights. The highest weights are set for the best spatial resolution. In this way, the cubes with the largest footprint would marginally contribute the final mosaics. The advantage of having this type of mosaic is to preserve the spectral information from the cubes acquired by VIMS with the best spatial resolution.

Mosaics characteristics:

- Cylindrically projected
- longitude range 360°W- 0°W
- latitude range -90° – +90°
- Spatial resolution for the projected hyperspectral cubes and the mosaics of each satellites set up to be 1 pixel/degree.

The first step to create the final mosaic is to cylindrically-project each VIMS-IR hyperspectral cube on a common surface, divided in bins of 1°longitude x 1°latitude by using the calculated geometric information.

Then, the cubes of each dataset were averaged, or weighted averaged, together where they spatially overlap. The weighted average is calculated with the frequency of spectra covering each pixel of the mosaic. The other non-weighted, is a regular average.

The file names indicate which processing steps have been applied:

SatelliteName_NormalizationStatus_Processing1_Processing2_Processing3_VersionNumber

Example: Rhea_notnorm_PC_w_modified_2, where:

- “notnorm” means the reflectance spectra were not normalized,
- “PC” stands for photometrically-corrected,
- “w” indicates the weighted average has been calculated,
- “modified” is for internal reference only.